

Multiplexed Network Formation*

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January 21, 2026

Abstract

We study a non-cooperative game in which individuals participate in multiple costly activities they have preferences over, and choose how much of their limited time to spend on each activity. This gives rise to an endogenously formed multiplexed network. We first provide conditions under which multiplexed networks naturally emerge, and under which they are socially efficient. We then show that agents are better off when preferences, costs and strategic complementarity parameters are heterogeneous rather than homogeneous across activities. We also provide a cautionary note on applying Bonacich centrality in multiplexed networks, which shows that the positive relationship between centrality and utility that holds in simplex networks does not necessarily carry over to multiplexed networks.

JEL classification: C72, D85

Keywords: Multiplexed networks, network formation, strategic complementarity and substitutability

*I am very grateful to my PhD supervisor, Markus Kinaterer, for his invaluable guidance and support throughout this research. I also extend my thanks to my Master thesis supervisors, Ugo Bolletta and Margherita Comola, for laying the foundation of my PhD. I am also very grateful to Francis Bloch and Kenan Huremović, as well as participants at 2023 BiNoMa and 2023 SAEe for helpful comments on this work. This work was supported by Asociación de Amigos de la Universidad de Navarra.

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1 Introduction

People share relationships that encompass interactions of different natures. They interact in multiple activities, such as lending money or exchanging information. The study of relationships in multiple activities has applications in trade and military alliances, crime and education, intergenerational transmission of inequality and product market competition (see [Joshi et al. 2025](#) for an overview). This multifaceted aspect of relationships has been empirically documented and shown to be an important feature in network formation ([Chandrasekhar et al., 2024](#); [Banerjee et al., 2023](#)).

The literature has defined the relationships in which individuals interact in only one activity (also called a layer in the literature) as *simplex*, and the ones in which individuals interact in more than one activity as *multiplexed*. Among the many models that aim to explain network formation, most consider only simplex networks. Multiplexed networks, such as the ones documented in [Chandrasekhar et al. \(2024\)](#) and [Banerjee et al. \(2023\)](#), have been mostly studied in other fields, such as sociology ([Corominas-Murtra et al., 2014](#)), physics ([De Domenico et al., 2013](#)) or computer science ([Cai et al., 2005](#)).

Our first contribution is to give a rationale for multiplexed networks. We study a setting in which multiplexed networks emerge endogenously from agents' time-allocation decisions across activities. This framework makes it possible to characterize when multiplexed networks arise in Nash equilibrium (Section 3.1) and when such multiplexed structures are socially efficient (Section 3.5). In the existing literature on multiplexed network formation, it is common to model interactions using games with linear-quadratic utility functions (see, for example, [Joshi et al. \(2025, 2020\)](#)). These utility functions are particularly tractable because Nash equilibrium values of agents correspond to their Bonacich centrality. Network games with linear-quadratic utility functions typically feature three key parameters: preferences for activities, costs of performing activities, and a level of strategic complementarity for each activity. However, these parameters are usually assumed to be homogeneous across activities.¹ In reality, not all activities share the

¹Some papers, such as [Chen et al. \(2018\)](#), consider heterogeneity in one or two of these

same characteristics, making heterogeneity an important and realistic dimension to consider. Our second contribution is to study how introducing heterogeneity in these parameters affects overall welfare (Sections 3.2, 3.3 and 3.4).

We study a network formation game à la [Cabrales et al. \(2011\)](#), where individuals allocate a limited amount of time between multiple types of activities and choose the weight of the link they form in each of these activities. Weights of links are determined by the equilibrium values of the game, so that the network is formed endogenously. In this type of network formation game, "the exact identity of the interacting partner is not an object of choice" ([Cabrales et al., 2011](#), p. 342); instead, what matters is the intensity of the link weights: network formation arises through the adjustment of link weights rather than through the selection of neighbors.

Individuals have heterogeneous preferences over activities, and participating in activities is costly. The cost of time allocation is convex and captures diminishing returns to intensity within a given activity, such as fatigue or cognitive overload. Agents derive utility both from their intrinsic preference for activities and from the enjoyment of interacting with others. Interdependencies across activities arise from the time constraint, which links agents' allocation decisions across layers. As shown in the remainder of the paper, preferences and strategic complementarities across agents drive the formation of multiplexed networks, while convex costs shape the intensity with which individuals interact in activities.

In the main analysis of this paper, we first provide conditions under which multiplexed networks naturally emerge, and conditions under which such multiplexed structures are socially efficient. We then show that heterogeneity in preferences for activities, in costs of performing activities, and in the level of strategic complementarity in activities increases welfare.

Throughout most of the analysis, we focus on parameter configurations under which the time constraint does not bind at equilibrium. This allows us to isolate how heterogeneity across activities affects equilibrium behavior and welfare in a tractable way. Section 3.4 then explicitly considers the case in which the time constraint binds and discusses which results remain robust.

parameters, and do not focus on the relationship between parameter heterogeneity and welfare.

In the section which covers the additional results of the paper, we show that one should be cautious when applying Bonacich centrality in multiplexed networks. In the existing literature, becoming more central in one activity always implies becoming better off, e.g. [Joshi et al. \(2025, 2020\)](#) or [Harkins \(2020\)](#). When applying Bonacich centrality in multiplexed networks, the relationship between utility and Bonacich centrality breaks.

Our paper is also related to [Baumann \(2021\)](#), who studies a [game of weighted network formation in which agents allocate a limited resource across relationships and self-investment](#). While both frameworks feature weighted network formation and resource constraints, the object of choice differs. In [Baumann \(2021\)](#), agents directly choose how much to invest in each relationship within a single-activity network. In contrast, in our model agents allocate time across multiple activities, and network links emerge endogenously from joint participation in these activities. This distinction allows us to study multiplexed networks and cross-activity trade-offs that are absent in single-activity models of network formation.

This paper is also closely related to the recent work of [Zenou and Zhou \(2025\)](#), who study network games on multiplexed networks under a resource constraint across activities. Their analysis takes the network structure as given and focuses on equilibrium characterization and policy in that environment. In contrast, the current paper focuses on how heterogeneity in preferences, costs, and strategic complementarities across activities affects welfare. While some of the behavioral mechanisms may also be present in models with exogenously given networks, our contribution is to isolate the welfare effects of parameter heterogeneity, which are not the object of analysis in [Zenou and Zhou \(2025\)](#). This work also relates to [Billand et al. \(2023\)](#) who study multiplexed network formation and consider a very general utility function. Differently from them, we study how heterogeneity in the game parameters affects welfare.²

The remainder of the paper is organized as follows. Section 2 presents the

²Our paper also relates to other recent works which study multiplexed networks that are not endogenous, or endogenous multiplexed networks that focus on other topics such as alliances and enmities between agents, or sustaining cooperation ([Mauleon et al., 2025](#); [Devine et al., 2024](#); [Kobayashi and Onaga, 2023](#); [Kor and Zhou, 2023](#); [Kundu and Pandey, 2023](#); [Tzavellas, 2023](#); [Cheng et al., 2021](#); [Walsh, 2022, 2019](#)).

model. Section 3 presents the results. Section 4 concludes.

2 The model

We next present the model, and introduce notation. We will define additional notation throughout the paper when necessary.

Activities and utility function. We consider a set $\mathcal{N} = \{1, \dots, N\}$ of agents who have access to a set $\mathcal{K} = \{1, \dots, K\}$ of activities, where $N, K \geq 2$. Each agent $i \in \mathcal{N}$ chooses to spend an amount of time $t_{i,k}$ performing activity $k \in \mathcal{K}$. We define set $\mathbf{t}_i = \{t_{i,1}, \dots, t_{i,K}\}$ as the time spent by agent i on each of the activities $k = 1, \dots, K$, and set $\mathbf{t}_{-i} = \{\mathbf{t}_1, \dots, \mathbf{t}_N\} \setminus \mathbf{t}_i$ as the time spent by all agents other than i on each activity. Each agent also has an exogenously given preference parameter $a_{i,k} \geq 0$ for activity k , which measures how much agent i enjoys performing activity k . Large values of $a_{i,k}$ indicate a high enjoyment of agent i for activity k . The preference parameter only captures the enjoyment of an agent for an activity, say practicing sports, and does not capture the enjoyment brought by spending time with other individuals who also perform the same activity, like playing football together. We assume that agents enjoy practicing at least one activity—i.e. $\sum_{k=1}^K a_{i,k} > 0$ for all $i \in \mathcal{N}$.

The utility brought to agent $i \in \mathcal{N}$ by spending time with individual $j \in \mathcal{N}$ on activity k , where $j \neq i$, is captured by the product $\alpha(t_{i,k}t_{j,k})$, where $t_{i,k} \in \mathbb{R}_+$ and $t_{j,k} \in \mathbb{R}_+$ are endogenously determined, as explained below. Parameter $\alpha > 0$ measures how much agents enjoy spending time together. It can be interpreted as a taste for socializing. Agents i and j will largely benefit from their interaction in activity k if (i) they both spend a large amount of time performing activity k , and (ii) they greatly enjoy social interactions. Agents gain utility by interacting with all the other agents of set $\mathcal{N}$, across each of the activities of set $\mathcal{K}$.

While participating in activities brings utility, it also requires effort. These costs of effort are measured by parameter $c > 0$. We represent the loss of utility brought to agent i by costs of performing activity k as convex in the time agent i spends in performing activity k . With such convex costs $c \cdot t_{i,k}^2$, agents have a decreasing incentive to spend an additional amount of time in performing activity k as the

amount of time they perform activity k —out of the limited amount of time they allocate between all activities—increases.

To summarize, there are three channels which determine the utility of agents. First, agents derive utility by **performing activities they enjoy**. Second, they derive utility by **spending time with other individuals who are performing the same activities**. Third, **performing activities is costly and costs increase in a convex way**. These forces are represented in the utility of agent $i \in \mathcal{N}$, which is written in (1):

$$U_i(\mathbf{t}_i, \mathbf{t}_{-i}) = \sum_{k=1}^K a_{i,k} t_{i,k} + \alpha \left(\sum_{k=1}^K \sum_{j \neq i} t_{i,k} t_{j,k} \right) - c \sum_{k=1}^K t_{i,k}^2, \quad (1)$$

given α , c and $a_{i,k}$ for all $i \in \mathcal{N}$ and all $k \in \mathcal{K}$. Note that $U_i : \mathbb{R}^{K \times N} \Rightarrow \mathbb{R}$ is a continuous and differentiable function.

Equilibrium. We denote by $\mathbf{t}_i^* = \{t_{i,1}^*, \dots, t_{i,K}^*\}$ the set of equilibrium values of agent i , which maximizes the utility of agent i . The set $\mathbf{t}^* = \{\mathbf{t}_1^*, \dots, \mathbf{t}_N^*\}$ is the Nash equilibrium of the game, in which no agent has a profitable unilateral deviation from her equilibrium value. If we let $\mathbf{t}'_i$ be any set $\mathbf{t}_i$, and $\mathbf{t}_{-i}^*$ be the set of equilibrium values of all agents other than i , then the Nash equilibrium is such that, for all agents $i \in \mathcal{N}$, $U_i(\mathbf{t}_i^*, \mathbf{t}_{-i}^*) \geq U_i(\mathbf{t}'_i, \mathbf{t}_{-i}^*)$ for all $\mathbf{t}'_i$. We denote by $\mathbf{U}$ the column vector in which entry i corresponds to the utility of agent i at equilibrium, denoted by U_i^* . We denote by $\mathbf{T}_k$ the column vector in which entry i corresponds to $t_{i,k}^*$.

Network formation process. The equilibrium values give rise to a weighted, undirected network in which agents are linked to each other in a maximum of K activities. The weight of the link between agents i and j in activity k , denoted by w_{ij}^k , takes a value defined as:

$$w_{ij}^k(\mathbf{t}^*) = t_{i,k}^* t_{j,k}^*. \quad (2)$$

We consider that a link in activity k is formed between agents i and j when $w_{ij}^k > 0$. The network formation process is essentially identical to that in [Cabrales](#)



Figure 1a: Simplex relationship.

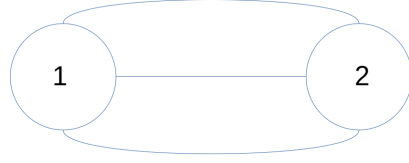


Figure 1b: Multiplexed relationship.

et al. (2011). Here, the focus of the network formation game is not on the identity of neighbors, but on the intensity of links with them, and on agents' decisions regarding which layers/activities to participate in.

Network definitions. We denote by $\mathbf{G}_k$ the $N \times N$ matrix such that entry ij takes value 1 if agents i and j share a link in activity k —i.e. $w_{ij}^k > 0$, and 0 otherwise. It can be thought of as the unweighted adjacency matrix of activity k . We define $\mathbf{G}_M = \sum_{k=1}^K \mathbf{G}_k$, which can be thought of as the unweighted matrix of the multiplexed network.

Matrix notation. Notation $\mathbf{X} \geq \mathbf{Y}$ ($\mathbf{X} > \mathbf{Y}$) means that each entry ij of matrix $\mathbf{X}$ is greater than or equal to (strictly greater than) the corresponding entry ij of matrix $\mathbf{Y}$. We denote by $\mathbf{1}$ a vector of ones, by $\mathbf{0}$ a vector of zeros, and by $\mathbf{I}_N$ the identity matrix of dimension N .

Economies. Throughout the paper, we use the term “economy” to refer to a given configuration of parameters α , c and $a_{i,k}$ for all $i \in \mathcal{N}$ and all $k \in \mathcal{K}$. To indicate a set for a specific configuration of parameters $\mathcal{E}$, we use superscript $\mathcal{E}$ on the corresponding vectors or parameters; that is, to indicate $\mathbf{U}$, $\mathbf{G}_M$, $a_{i,k}$ and U_i^* in economy $\mathcal{E}$, they are written as $\mathbf{U}^{\mathcal{E}}$, $\mathbf{G}_M^{\mathcal{E}}$, $a_{i,k}^{\mathcal{E}}$ and $U_i^{*\mathcal{E}}$ respectively.

Multiplexity. We say that individuals share a multiplexed relationship if they form a link in two or more activities. Formally, the multiplexity of a relationship is defined in Definition 1.

Definition 1 (Multiplexity). *The multiplexity MX_{ij} of a relationship between agents i and j corresponds to the entry ij of $\mathbf{G}_M$.*

The multiplexity of a relationship between two agents is simply the number of links they share. For instance, the multiplexity of the relationship between agents 1 and 2 in the network of Figure 1a is 1, whereas it is 3 in the network of Figure 1b. We consider that a network fulfills multiplexity, or that it is multiplexed, if there exist two agents i and j such that $MX_{ij} \geq 2$, and that it is simplex if it does not fulfill multiplexity.

3 Main results

3.1 Existence of equilibrium and multiplexity

We study a linear–quadratic game in which agents allocate time across multiple activities. Throughout the analysis, we focus on interior equilibria in which agents devote at most one unit of time to activities. This restriction ensures that equilibrium time allocations are economically meaningful and that link weights remain in the interval $[0, 1]$.

Each agent $i \in \mathcal{N}$ solves maximization problem (3) by choosing to spend an amount of time $t_{i,k} \in \mathbb{R}_+$ on each activity $k \in \mathcal{K}$:

$$\max_{\mathbf{t}_i} U_i(\cdot) = \sum_{k=1}^K a_{i,k} t_{i,k} + \alpha \left(\sum_{k=1}^K \sum_{j \neq i} t_{i,k} t_{j,k} \right) - c \sum_{k=1}^K t_{i,k}^2, \quad (3)$$

and we restrict our attention to interior solutions, i.e. agents spend less than one unit of time across activities. In Section 3.4, we study the case in which solutions are not necessarily interior, i.e. agents spend all their time across activities. The equilibrium values of agents are given by the system of best-response functions defined in (4) for all $k \in \mathcal{K}$.

$$\begin{cases} t_{1,k} = \frac{a_{1,k} + \alpha(\sum_{j \neq 1} t_{j,k})}{2c}, \\ \dots \\ t_{N,k} = \frac{a_{N,k} + \alpha(\sum_{j \neq N} t_{j,k})}{2c}. \end{cases} \quad (4)$$

The optimal amount of time agent i spends on activity k is increasing in her preference for activity k and the amount of time agents other than her spend

on activity k , and decreasing in the cost of spending time on activities. It is worth noting from (4) that the decision of agents to spend time on activities depends solely on their preferences and on the benefits that stem from strategic complementarities. Costs affect the intensity of participation, but do not determine which activities agents engage in. This idea will be formally stated in Lemma 1 below.

The Nash equilibrium in activity k is given by vector $\mathbf{T}_k$, composed of one column and N rows.

$$\mathbf{T}_k = [2c \cdot \mathbf{I}_N - \alpha \cdot \mathbf{\Psi}]^{-1} \mathbf{A}_k^\top, \quad (5)$$

where $\mathbf{I}_N$ is the identity matrix of dimensions N , while in matrix $\mathbf{\Psi}$, of dimensions $N \times N$, all entries in the main diagonal are zero's, and all other entries are one's, and $\mathbf{A}_k$ is a $1 \times N$ row vector given by:

$$\mathbf{A}_k = \begin{bmatrix} a_{1,k} & \cdots & a_{N,k} \end{bmatrix}.$$

We make the following assumption to ensure the existence of a unique Nash equilibrium, in which equilibrium values are non-negative.

Assumption 1. $\alpha < \frac{2c}{n-1}$.

Assumption 1 is very standard in games with a linear-quadratic utility function, and ensures that the equilibrium in (5) exists and is unique. For illustrative purposes, we next provide an example of how the network forms.

Example 1. The parameters of preferences are given in Table 1. We set costs of performing activities to $c = \frac{3}{2}$, and $\alpha = 1$. The resulting equilibrium values are given in Table 2. Given the parameter values, the equilibrium values, and equation (2), links' weights take the values presented in Table 3 and the network represented in Figure 2 arises. $\square$

We first find that agents will spend time on activities as long as they enjoy practicing them, or at least one other agent enjoys practicing them. Even though agents do not enjoy certain activities, they are better off spending time on them in or-

Preference parameters			
	Agent 1	Agent 2	Agent 3
Activity R	$a_{1,R} = 0.8$	$a_{2,R} = 0.5$	$a_{3,R} = 0.4$
Activity S	$a_{1,S} = 0.1$	$a_{2,S} = 0.4$	$a_{3,S} = 0.3$
Activity T	$a_{1,T} = 0.1$	$a_{2,T} = 0.1$	$a_{3,T} = 0.3$

Table 1: Preference parameters.

Equilibrium values			
	Agent 1	Agent 2	Agent 3
Activity R	$t_{1,R}^* = 0.625$	$t_{2,R}^* = 0.55$	$t_{3,R}^* = 0.525$
Activity S	$t_{1,S}^* = 0.225$	$t_{2,S}^* = 0.3$	$t_{3,S}^* = 0.275$
Activity T	$t_{1,T}^* = 0.15$	$t_{2,T}^* = 0.15$	$t_{3,T}^* = 0.2$

Table 2: Equilibrium values given the preference parameters in Table 1, $c = \frac{3}{2}$ and $\alpha = 1$.

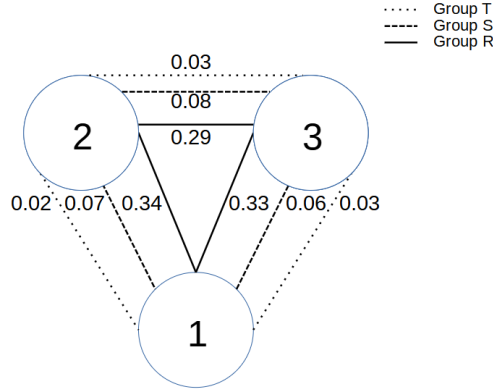


Figure 2: Weighted multiplexed network for weights of links in Table 3.

der to interact with other agents, and benefit from strategic complementarities. Hence, a network fulfills multiplexity if and only if there exist two activities that are enjoyed by at least one agent each. It is worth noting that the emergence of multiplexity follows directly from the preferences of agents and the utility they can derive by interacting with other agents, while convex costs shape the intensity of participation in activities.

Lemma 1. *Under Assumption 1,*

Weights of links			
	w_{12}	w_{13}	w_{23}
Activity R	$w_{12}^R = 0.34$	$w_{13}^R = 0.33$	$w_{23}^R = 0.29$
Activity S	$w_{12}^S = 0.07$	$w_{13}^S = 0.06$	$w_{23}^S = 0.08$
Activity T	$w_{12}^T = 0.02$	$w_{13}^T = 0.03$	$w_{23}^T = 0.03$

Table 3: Weights of links given equilibrium values in Table 2.

- (i) an agent i spends time on an activity k she does not enjoy ($a_{i,k} = 0$) if and only if at least another agent enjoys activity k ($\exists j$ such that $a_{j,k} > 0$), and
- (ii) the network fulfills multiplexity if and only if there exist two activities k and l , with $k \neq l$, such that $a_{i,k} > 0$ and $a_{j,l} > 0$ for any $i, j \in \mathcal{N}$.

On the contrary, when all agents enjoy practicing the same activity, and do not enjoy any other activity, they will spend all their time practicing that activity. This equilibrium will lead agents to form a simplex network, in which they are only connected in the activity they enjoy.

Corollary 1. *Under Assumption 1, the network is simplex if and only if all agents $i \in \mathcal{N}$ are such that $a_{i,k} > 0$ for one activity $k \in \mathcal{K}$, and $a_{i,l} = 0$ for all other activities $l \neq k$.*

Lemma 1 implies that, under our assumptions, networks within each activity are complete whenever at least one agent strictly enjoys that activity. This feature is a direct consequence of strategic complementarities and reflects the fact that agents may participate in activities primarily to interact with others rather than for intrinsic enjoyment. Importantly, the focus of the paper is not on explaining which specific pairs of agents interact within a given activity, but on how agents allocate time across multiple activities and how this generates multiplexed network structures. Allowing for negative preferences or fixed costs of link formation could prevent some agents from participating in certain activities, but such extensions are beyond the scope of this paper. As shown in Section 3.5 below, the condition for multiplexed networks to arise in equilibrium (Lemma 1) differs from the condition for such networks to be socially efficient.

3.2 Welfare and heterogeneity in parameters $a_{i,k}$

We next show that agents have a larger utility when they have heterogeneous preferences than when they have homogeneous preferences, in Proposition 1 below. The size of the spread in preference parameters determines the size of the increase in utility. Before formally stating Proposition 1 below, we define what an economy of *homogeneous preferences* and an economy of *heterogeneous preferences* are, and provide some examples that illustrate the idea of Proposition 1.

Among the many possible ways to introduce heterogeneity across activities, we focus on mean-preserving spreads. This restriction allows us to study the welfare effects of heterogeneity per se, rather than changes driven by uniform shifts in the overall attractiveness of activities. When the mean of preference parameters is held constant, any welfare variation arises from differences in how agents allocate effort across activities, rather than from an increase in the total time spent across activities. Mean-preserving spreads therefore provide a natural way to isolate the impact of heterogeneity on equilibrium behavior and welfare.

Heterogeneity in preferences. We say that economy $\mathcal{E}$ is one of *homogeneous preferences* if $a_{i,k} = a_{j,k}$ for all $i, j \in \mathcal{N}$ and all $k \in \mathcal{K}$. Given any economy of homogeneous preferences $\mathcal{E}$, we say that economy $\mathcal{E}'$ is an economy of *heterogeneous preferences*, if (i) $\frac{\sum_{k=1}^K a_{i,k}}{K}$ is the same in both economies $\mathcal{E}'$ and $\mathcal{E}$ for all $i \in \mathcal{N}$, and (ii) $a_{i,k}^{\mathcal{E}} \geq a_{i,l}^{\mathcal{E}}$ implies $a_{i,k}^{\mathcal{E}'} \geq a_{i,k}^{\mathcal{E}}$ and $a_{i,l}^{\mathcal{E}'} \leq a_{i,l}^{\mathcal{E}}$ for all $i \in \mathcal{N}$ and all $k, l \in \mathcal{K}$, and $a_{i,k}^{\mathcal{E}'} > a_{i,k}^{\mathcal{E}}$ and $a_{i,l}^{\mathcal{E}'} < a_{i,l}^{\mathcal{E}}$ for at least one agent $i \in \mathcal{N}$ and at least two activities $k, l \in \mathcal{K}$. Condition (ii) ensures that there is a spread in preferences in economy $\mathcal{E}'$ relative to economy $\mathcal{E}$, and condition (i) that it is mean-preserving. In the remainder of the paper, we study mean-preserving economies as defined above, and relax condition (i) in Appendix A below. Given any economy of homogeneous preferences $\mathcal{E}$ and any two economies of heterogeneous preferences $\mathcal{E}'$ and $\mathcal{E}''$, we say that economy $\mathcal{E}''$ has a *larger spread* than $\mathcal{E}'$ with respect to economy $\mathcal{E}$ if (i) $\frac{\sum_{k=1}^K a_{i,k}}{K}$ is the same in the three economies $\mathcal{E}''$, $\mathcal{E}'$ and $\mathcal{E}$ for all $i \in \mathcal{N}$, and (ii) $a_{i,k}^{\mathcal{E}'} \geq a_{i,l}^{\mathcal{E}'}$ implies $a_{i,k}^{\mathcal{E}''} \geq a_{i,k}^{\mathcal{E}'}$ and $a_{i,l}^{\mathcal{E}''} \leq a_{i,l}^{\mathcal{E}'}$ for all $i \in \mathcal{N}$ and all $k, l \in \mathcal{K}$, and $a_{i,k}^{\mathcal{E}''} > a_{i,k}^{\mathcal{E}'}$ and $a_{i,l}^{\mathcal{E}''} < a_{i,l}^{\mathcal{E}'}$ for at least one agent $i \in \mathcal{N}$ and at least two activities $k, l \in \mathcal{K}$. In other words, economy $\mathcal{E}''$ has a larger spread than economy $\mathcal{E}'$ with

respect to economy $\mathcal{E}$, if $\mathcal{E}''$ corresponds to a mean-preserving spread in preferences of $\mathcal{E}'$, and $\mathcal{E}'$ corresponds to a mean-preserving spread in preferences of $\mathcal{E}$. An economy of heterogeneous preferences $\mathcal{E}'$ is *symmetric* if, for every preference parameter $a_{i,k}^{\mathcal{E}'} = a_{i,k}^{\mathcal{E}} + \phi_{i,k}$ given some $\phi_{i,k} \in \mathbb{R}$, there exists exactly one preference parameter $a_{j,k}^{\mathcal{E}'} = a_{j,k}^{\mathcal{E}} - \phi_{i,k}$. We next give an example of a symmetric economy of heterogeneous preferences $\mathcal{E}'$, presented in Table 5, and an economy of homogeneous preferences $\mathcal{E}$, presented in Table 4.

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 0.5$	$a_{2,R} = 0.5$	$a_{3,R} = 0.5$	$a_{4,R} = 0.5$
Activity S	$a_{1,S} = 0.5$	$a_{2,S} = 0.5$	$a_{3,S} = 0.5$	$a_{4,S} = 0.5$

Table 4: Economy $\mathcal{E}$

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 0.9$	$a_{2,R} = 0.7$	$a_{3,R} = 0.3$	$a_{4,R} = 0.1$
Activity S	$a_{1,S} = 0.1$	$a_{2,S} = 0.3$	$a_{3,S} = 0.7$	$a_{4,S} = 0.9$

Table 5: Economy $\mathcal{E}'$

The spread of economy $\mathcal{E}'$ with respect to economy $\mathcal{E}$ is mean-preserving, since in both economies $\mathcal{E}'$ and $\mathcal{E}$, the mean $\frac{\sum_{k=1}^K a_{i,k}}{2} = 0.5$ is the same for all agents. It is also symmetric, since, for every parameter $a_{i,R} = 0.5 + \phi_{i,R}$ ($a_{i,S} = 0.5 + \phi_{i,S}$), there is exactly one parameter $a_{j,R} = 0.5 - \phi_{i,R}$ ($a_{j,S} = 0.5 - \phi_{i,S}$), where $\phi_{i,R} \in \{0.4, 0.2\}$ ($\phi_{i,S} \in \{0.4, 0.2\}$). If there were, for example, exactly one agent for whom $a_{i,R} = 0.5 + \phi_{i,R}$, and exactly two agents j, l for whom $a_{m,R} = 0.5 - \phi_{i,R}$, where $m \in \{j, l\}$, then economy $\mathcal{E}'$ would not be symmetric. We next give an example of an economy of heterogeneous preferences, that has a larger spread than $\mathcal{E}'$ with respect to $\mathcal{E}$, in Table 6.

We now illustrate, by means of two examples, that agents who are in an economy of heterogeneous preferences $\mathcal{E}'$ have a larger utility than in an economy of

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 0.9$	$a_{2,R} = 1$	$a_{3,R} = 0$	$a_{4,R} = 0.1$
Activity S	$a_{1,S} = 0.1$	$a_{2,S} = 0$	$a_{3,S} = 1$	$a_{4,S} = 0.9$

Table 6: Economy $\mathcal{E}''$

homogeneous preferences $\mathcal{E}$. We consider three economies: one of homogeneous preferences, which is displayed above in Table 4, one of heterogeneous preferences in Table 7, and a second one of heterogeneous preferences in Table 8, where the spread is larger. The purpose of Example 2 is to illustrate that agents are better off in economies of heterogeneous preferences, and to provide intuition for the result that utility increases with the size of the preference spread. Heterogeneity improves welfare through specialization: agents spend most of their time on the activities they benefit the most from, and specialization benefits increase with the degree of specialization.

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 0.5$	$a_{2,R} = 0.75$	$a_{3,R} = 0.25$	$a_{4,R} = 0.5$
Activity S	$a_{1,S} = 0.5$	$a_{2,S} = 0.25$	$a_{3,S} = 0.75$	$a_{4,S} = 0.5$

Table 7: Heterogeneous preferences

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 0.5$	$a_{2,R} = 1$	$a_{3,R} = 0$	$a_{4,R} = 0.5$
Activity S	$a_{1,S} = 0.5$	$a_{2,S} = 0$	$a_{3,S} = 1$	$a_{4,S} = 0.5$

Table 8: Heterogeneous preferences

Example 2. Suppose that $c = 2$ and $\alpha = 1$, and consider the preference parameters given in Table 4. Then,

$$\mathbf{T}_R = \mathbf{T}_S = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

Suppose that $c = 2$ and $\alpha = 1$, and consider the preference parameters given in Table 7. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.5 \\ 0.55 \\ 0.45 \\ 0.5 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.5 \\ 0.45 \\ 0.55 \\ 0.5 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 1 \\ 1.01 \\ 1.01 \\ 1 \end{bmatrix}.$$

Suppose that $c = 2$ and $\alpha = 1$, and consider the preference parameters given in Table 8. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.5 \\ 0.6 \\ 0.4 \\ 0.5 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.5 \\ 0.4 \\ 0.6 \\ 0.5 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 1 \\ 1.04 \\ 1.04 \\ 1 \end{bmatrix}.$$

Agents prefer the economy of Table 8 to the one of Table 7, which they in turn prefer to the one of Table 4. $\square$

More generally, agents are better off in economies of heterogeneous preferences than in economies of homogeneous preferences. This happens because agents derive a large amount of utility by practicing activities they particularly enjoy, even if it entails spending less time with other individuals. Preference heterogeneity allows agents to specialize in the activities they value most, reallocating time away from less valued activities and thereby increasing welfare through more efficient specialization. Example 2 is not meant to be exhaustive, but to make transparent the specialization mechanism underlying Proposition 1. We next show that this result is robust to economies of heterogeneous preferences other than the ones presented in Table 7 and 8, such as the two we consider next: one in which there is spread in the preferences of more than two agents, in Table 9, and one economy of heterogeneous preferences which is not symmetric, in Table 10.

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 1$	$a_{2,R} = 1$	$a_{3,R} = 0$	$a_{4,R} = 0$
Activity S	$a_{1,S} = 0$	$a_{2,S} = 0$	$a_{3,S} = 1$	$a_{4,S} = 1$

Table 9: Heterogeneous preferences

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 1$	$a_{2,R} = 0$	$a_{3,R} = 0$	$a_{4,R} = 0$
Activity S	$a_{1,S} = 0$	$a_{2,S} = 1$	$a_{3,S} = 1$	$a_{4,S} = 1$

Table 10: Heterogeneous preferences

Example 3. Suppose that $c = 2$ and $\alpha = 1$, and consider the preference parameters given in Table 9. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.6 \\ 0.6 \\ 0.4 \\ 0.4 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.4 \\ 0.4 \\ 0.6 \\ 0.6 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 1.04 \\ 1.04 \\ 1.04 \\ 1.04 \end{bmatrix}.$$

Suppose that $c = 2$ and $\alpha = 1$, and consider the preference parameters given in Table 10. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.4 \\ 0.2 \\ 0.2 \\ 0.2 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.6 \\ 0.8 \\ 0.8 \\ 0.8 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 1.04 \\ 1.36 \\ 1.36 \\ 1.36 \end{bmatrix}.$$

Agents prefer the economy of Table 10 to the one of Table 9, which they in turn prefer to the one of Table 4. $\square$

Some of the conclusions we can draw from specific examples of Tables 9 and 10 are general, as we show formally in Proposition 1 below. First, agents are better off in economies of heterogeneous preferences than in economies of homogeneous preferences, regardless of the number of agents that the spread affects. Second, the economy of heterogeneous preferences does not need to be symmetric for agents to

have a larger utility than in an economy of homogeneous preferences. However, an economy of heterogeneous preferences with a larger spread implies larger utilities of agents only if both economies of heterogeneous preferences are symmetric. To improve the readability of this paper, we do not include examples of economies of heterogeneous preferences with spreads over 3 or more activities, but in these cases, which Proposition 1 below accounts for, agents are also better off than in an economy of homogeneous preferences.

Proposition 1. *Suppose that $\mathcal{E}$ is an economy of homogeneous preferences. Under Assumption 1,*

- (i) *all agents have a weakly larger utility in an economy of heterogeneous preferences $\mathcal{E}'$ than in $\mathcal{E}$ ($U_i^{*\mathcal{E}'} \geq U_i^{*\mathcal{E}}$ for all $i \in \mathcal{N}$), and*
- (ii) *for any two symmetric economies of heterogeneous preferences $\mathcal{E}'$ and $\mathcal{E}''$, all agents have a weakly larger utility in the economy of heterogeneous preferences that has the larger spread with respect to $\mathcal{E}$ (if $\mathcal{E}''$ has a larger spread than $\mathcal{E}'$, then $U_i^{*\mathcal{E}''} \geq U_i^{*\mathcal{E}'}$ for all $i \in \mathcal{N}$).*

A natural follow-up question to whether heterogeneity raises welfare concerns inequalities. To what extent do heterogeneous economies create inequalities? Agents with spread-out preferences specialize in the activities they benefit the most from, leading to a higher individual utility. By contrast, agents who keep homogeneous preferences do not benefit from gains in utility, resulting in inequalities between the agents who have heterogeneous preferences and those have homogeneous ones. Larger spreads in preferences amplify inequalities. When spreads are not mean-preserving, inequalities arise not only from changes in agents' time allocations across activities, but also from changes in the total amount of time they allocate to those activities. In general, agents who increase their total time spent across activities also derive higher utility.

3.3 Welfare and heterogeneity in parameters α_k and c_k

Up to now, we have considered that costs c and taste for socializing α are both homogeneous across activities. However, it is realistic to assume that both α and c

are heterogeneous across activities. In this case, the utility function of each agent i is:

$$U_i(\mathbf{t}_i, \mathbf{t}_{-i}) = \sum_{k=1}^K a_{i,k} t_{i,k} + \sum_{k=1}^K \sum_{j \neq i} \alpha_k t_{i,k} t_{j,k} - \sum_{k=1}^K c_k t_{i,k}^2. \quad (6)$$

In this framework, we need to impose a different condition on the threshold of α that accounts for heterogeneity in α_k and c_k .

Assumption 2. $\alpha_k < \frac{2c_k}{n-1}$ for all $k \in \mathcal{K}$.

In the same spirit as economies of heterogeneous preferences $a_{i,k}$ considered above, we consider economies of *heterogeneous parameters* α_k and c_k , and show that agents are also better off when parameters α_k and c_k are heterogeneous across activities than when they are homogeneous. We first formally define an economy of *homogeneous parameters* and an *economy of heterogeneous parameters* α_k (or c_k). Then, we provide some examples that illustrate that heterogeneity in parameters α_k and c_k increases welfare, and formally state the result in Proposition 2 below. It is worth noting an important distinction with respect to heterogeneity in preference parameters. While mean-preserving spreads in preference parameters isolate pure reallocation effects, mean-preserving spreads in costs and strategic complementarities also affect the overall time that agents allocate across activities. In the next section, we study how welfare reacts to spreads in α_k and c_k when heterogeneity does not affect the total time that agents spend on activities.

Heterogeneity in parameters α_k and c_k . An economy $\mathcal{E}$ is one of homogeneous parameters if $\alpha_k = \alpha_l$, $c_k = c_l$ for all $k \in \mathcal{K}$, and $a_{i,k} = a_{j,l}$ for all $i, j \in \mathcal{N}$ and all $k, l \in \mathcal{K}$. An economy $\mathcal{E}'$ is an economy of heterogeneous parameters α_k (c_k) with respect to economy $\mathcal{E}$, if (i) $\frac{\sum_{k=1}^K \alpha_k}{K}$ ($\frac{\sum_{k=1}^K c_k}{K}$) is the same in both economies $\mathcal{E}'$ and $\mathcal{E}$, (ii) $\alpha_k^{\mathcal{E}} \geq \alpha_l^{\mathcal{E}}$ implies $\alpha_k^{\mathcal{E}'} \geq \alpha_l^{\mathcal{E}}$ and $\alpha_l^{\mathcal{E}'} \leq \alpha_l^{\mathcal{E}}$ for all $k, l \in \mathcal{N}$, and $\alpha_k^{\mathcal{E}'} > \alpha_k^{\mathcal{E}}$ and $\alpha_l^{\mathcal{E}'} < \alpha_l^{\mathcal{E}}$ for at least two activities $k, l \in \mathcal{K}$ ($c_k^{\mathcal{E}} \geq c_l^{\mathcal{E}}$ implies $c_k^{\mathcal{E}'} \geq c_l^{\mathcal{E}}$ and $c_l^{\mathcal{E}'} \leq c_l^{\mathcal{E}}$ for all $k, l \in \mathcal{N}$, and $c_k^{\mathcal{E}'} > c_k^{\mathcal{E}}$ and $c_l^{\mathcal{E}'} < c_l^{\mathcal{E}}$ for at least two activities $k, l \in \mathcal{K}$), and (iii) $c_k = c_l$ ($\alpha_k = \alpha_l$) for all $k, l \in \mathcal{K}$ and $a_{i,k} = a_{j,l}$ for all $k, l \in \mathcal{K}$ and $i, j \in \mathcal{N}$. Given any economy $\mathcal{E}$ of homogeneous parameters and any two

economies $\mathcal{E}'$ and $\mathcal{E}''$ of heterogeneous parameters α_k (c_k), we say that economy $\mathcal{E}''$ has a larger spread than $\mathcal{E}'$ with respect to economy $\mathcal{E}$ if (i) $\frac{\sum_{k=1}^K \alpha_k}{K}$ ($\frac{\sum_{k=1}^K c_k}{K}$) is the same in the three economies $\mathcal{E}''$, $\mathcal{E}'$ and $\mathcal{E}$, (ii) $\alpha_k^{\mathcal{E}'} \geq \alpha_l^{\mathcal{E}'}$ implies $\alpha_k^{\mathcal{E}''} \geq \alpha_l^{\mathcal{E}'}$ and $\alpha_l^{\mathcal{E}''} \leq \alpha_l^{\mathcal{E}'}$ for all $k, l \in \mathcal{N}$, and $\alpha_k^{\mathcal{E}''} > \alpha_k^{\mathcal{E}'}$ and $\alpha_l^{\mathcal{E}''} < \alpha_l^{\mathcal{E}'}$ for at least two activities $k, l \in \mathcal{K}$ ((ii) $c_k^{\mathcal{E}'} \geq c_l^{\mathcal{E}'}$ implies $c_k^{\mathcal{E}''} \geq c_k^{\mathcal{E}'}$ and $c_l^{\mathcal{E}''} \leq c_l^{\mathcal{E}'}$ for all $k, l \in \mathcal{N}$, and $c_k^{\mathcal{E}''} > c_k^{\mathcal{E}'}$ and $c_l^{\mathcal{E}''} < c_l^{\mathcal{E}'}$ for at least two activities $k, l \in \mathcal{K}$), and (iii) $c_k = c_l$ ($\alpha_k = \alpha_l$) for all $k, l \in \mathcal{K}$ and $a_{i,k} = a_{j,l}$ for all $k, l \in \mathcal{K}$ and $i, j \in \mathcal{N}$.

We next show, by means of an example, that agents are better off in economies of heterogeneous parameters α_k .

Example 4. Suppose that $c = 3$ and $\alpha = 1$, and consider the preference parameters given in Table 4. Then,

$$\mathbf{T}_R = \mathbf{T}_S = \begin{bmatrix} 0.17 \\ 0.17 \\ 0.17 \\ 0.17 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 0.17 \\ 0.17 \\ 0.17 \\ 0.17 \end{bmatrix}.$$

Suppose that $c = 3$, $\alpha_R = 1.2$, $\alpha_S = 0.8$, and consider the preference parameters given in Table 4. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.21 \\ 0.21 \\ 0.21 \\ 0.21 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.14 \\ 0.14 \\ 0.14 \\ 0.14 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 0.19 \\ 0.19 \\ 0.19 \\ 0.19 \end{bmatrix}.$$

Agents have a larger utility in the economy in which the α_k parameters are heterogeneous. $\square$

In economies of preferences parameters α_k or c_k , agents are better off than in economies of homogeneous parameters. In this sense, the result of Proposition 1 is robust to heterogeneity in parameters other than preferences $a_{i,k}$.

Proposition 2. Suppose that $\mathcal{E}$ is an economy of homogeneous preferences. Under Assumption 2,

- (i) all agents have a strictly larger utility in an economy $\mathcal{E}'$ of heterogeneous parameters (α_k or c_k) than in $\mathcal{E}$ ($U_i^{\mathcal{E}'} > U_i^{\mathcal{E}}$ for all $i \in \mathcal{N}$), and
- (ii) for any two economies $\mathcal{E}'$ and $\mathcal{E}''$ of heterogeneous parameters (α_k or c_k), all agents have a strictly larger utility in the economy of heterogeneous parameters that has the larger spread with respect to $\mathcal{E}$ (if $\mathcal{E}''$ has a larger spread than $\mathcal{E}'$, then $U_i^{\mathcal{E}''} > U_i^{\mathcal{E}'}$ for all $i \in \mathcal{N}$).

Heterogeneity in α_k and c_k improves welfare through specialization across activities. When strategic complementarities are heterogeneous, agents concentrate their time on activities with large coordination benefits, due to strong spillovers in such activities. Similarly, when costs are heterogeneous, agents re-allocate their time away from costly activities to cheaper ones, reducing the marginal cost of participation. In both cases, heterogeneity creates comparative advantages across activities.

3.4 Time constraint, outside activity and corner solutions

Up until now, we have focused on interior solutions, i.e. cases in which the time constraint does not bind. In these cases, we have implicitly assumed that the remaining time that agents do not spend on social activities is relegated to a non-social, outside activity, such as passive resting. We now turn our attention to cases in which agents do not spend time on the outside activity, i.e. the time constraint binds.

Each agent $i \in \mathcal{N}$ solves maximization problem (7):

$$\begin{aligned} \max_{\mathbf{t}_i} \quad & U_i(\cdot) = \sum_{k=1}^K a_{i,k} t_{i,k} + \sum_{k=1}^K \sum_{j \neq i} \alpha_k t_{i,k} t_{j,k} - \sum_{k=1}^K c_k t_{i,k}^2 \\ \text{s.t.} \quad & \sum_{k=1}^K t_{i,k} + t_{i,0} = 1, \end{aligned} \quad (7)$$

where $t_{i,0}$ is the time that agent i spends on the outside activity. The corresponding Lagrangean is:

$$U_i(\cdot) = \sum_{k=1}^K a_{i,k} t_{i,k} + \sum_{k=1}^K \sum_{j \neq i} \alpha_k t_{i,k} t_{j,k} - \sum_{k=1}^K c_k t_{i,k}^2 + \lambda_i (1 - \sum_{k=1}^K t_{i,k} - t_{i,0}), \quad (8)$$

where $\lambda_i > 0$ since the time constraint is binding. After some algebra, one finds that the best-response function of agent i is:

$$t_{i,k} = \frac{a_{i,k} + \alpha_k \sum_{j \neq i} t_{j,k} - \lambda_i}{2c_k}, \quad (9)$$

and that the Nash equilibrium is given by:

$$\mathbf{T}_k = [2c_k - \alpha_k \mathbf{\Psi}]^{-1}(\mathbf{A}_k^\top - \boldsymbol{\lambda}), \quad (10)$$

where

$$\boldsymbol{\lambda} = \left(\sum_{k=1}^K [2c_k \mathbf{I}_N - \alpha_k \mathbf{\Psi}]^{-1} \right) \left(\sum_{k=1}^K [2c_k \mathbf{I}_N - \alpha_k \mathbf{\Psi}]^{-1} \mathbf{A}_k^\top - \mathbf{1} \right), \quad (11)$$

where entry i of $\boldsymbol{\lambda}$ corresponds to λ_i . Notice that the maximization problem that is solved in Section 3.1 corresponds to the maximization problem in (7) when $\lambda_i = 0$, and thus, the Nash equilibrium in Section 3.1 corresponds to the Nash equilibrium in (10) when $\boldsymbol{\lambda} = \mathbf{0}$.

When the time constraint binds, the condition given in Lemma 1 for the formation of multiplexed networks is not sufficient anymore. The best-response function in (9) shows that the agent must have a large enough preference for activity k and/or a large enough benefit from strategic complementarities in order to cover the shadow price of time.

The results on welfare and heterogeneity in preference parameters are robust to the introduction of a binding time constraint. Notice from (11) that, when costs and strategic complementarities are homogeneous, the value of λ_i is independent of the distribution of preference parameters. As a result, mean-preserving spreads in preference parameters induce exactly the same reallocation of time across activities as in the interior solution case. Since the welfare effects derived in interior solution case rely exclusively on these reallocation effects, all heterogeneity results remain unchanged when the time constraint binds. Mean-preserving spreads in strategic complementarities α_k also raise welfare, since they render the distribution of time across activities more concentrated on certain activities. Because the utility of agent i at equilibrium is given by $\sum_{k=1}^K c_k t_{i,k}^{*2}$, concentrating one's time on certain activities raises welfare. The effect of mean-preserving spreads in c_k on welfare are

less obvious: a concentration of agents' time on certain activities does not mechanically increase their utility $\sum_{k=1}^K c_k t_{i,k}^{*2}$ at equilibrium. When costs are already large enough relative to strategic complementarities, mean-preserving spreads in costs lower welfare, as concentrating one's time on activities becomes very costly. By contrast, when strategic complementarities are large enough relative to costs, the benefit from coordinating in certain activities with other agents offsets the convex costs of specialization, and thus, welfare increases. When the time constraint is not binding, mean-preserving spreads in costs always raise welfare because (i) costs are large enough relative to strategic complementarities, and (ii) mean-preserving spreads in costs raise the total amount of time that agents spend across activities.

3.5 A social planner benchmark

We now consider the case of a social planner whose objective function is to maximize the sum of utilities, and who can decide on how each agent i allocates a certain amount of time $\bar{T}_i$, independently of agents' preferences for activities. Because the social planner can determine the time agents allocate to activities, she can thereby decide on the structure of the network. Note as well that there is an infinity of possible allocations that can be achieved. This contrasts with the case where agents choose their own time allocation since the equilibrium, given in (5), is unique. We find that, if c is large enough, then the network structure which maximizes the sum of utilities is multiplexed, whereas it is simplex if c is low enough. This is the case because, if costs c are large enough, the disutility $c \cdot \sum_{k=1}^K t_{i,k}^2$ brought by spending time on only one activity is very large. Thus, the economy which maximizes the sum of utilities is one in which agents spend time on multiple activities, thereby forming a multiplexed network.

Proposition 3. *Given any set of parameters, there exists some threshold $\underline{c}$ such that, if $c > \underline{c}$, then the network structure preferred by the social planner is multiplexed.*

Threshold $\underline{c}$ takes a large value when agents all enjoy the same activity and α is large, since the simplex network yields a large sum of utilities, whereas it takes a

lower value when agents enjoy different activities and α is lower. A natural follow-up question to Proposition 3 is whether there exists a range of values of c below $\underline{c}$ for which a multiplexed and a simplex networks are both socially efficient. If there exists a value for which a multiplexed and a simplex network are both socially efficient, then it is unique. Suppose that there exists a value of c , denoted by c_1 , such that both a multiplexed and a simplex network are socially efficient, and, *ad absurdum*, that there exists a second value of c , denoted by c_2 , such that both a simplex and multiplexed network are also socially efficient. Because $\frac{\partial \sum_{i=1}^N U_i}{\partial c}$ is negative and larger in absolute value in the simplex network economy than in the multiplexed network, and $\frac{\partial^2 \sum_{i=1}^N U_i}{\partial c^2} = 0$, value c_2 does not exist, and a contradiction arises.

Corollary 2. *If $c < \underline{c}$, then the network structure preferred by the social planner is simplex.*

The network structure preferred by the social planner does not always coincide with the network structure that arises in equilibrium. Notice that when costs are low enough and agents enjoy practicing multiple activities, the structure preferred by the social planner is the simplex network, while the structure that arises at equilibrium is the multiplexed network. As explained in Section 3.1, agents spend some positive time on an activity as long as someone spends time on the activity, in order to benefit from strategic complementarities, regardless of whether costs are low enough or not.

3.6 A cautionary note on applying Bonacich centrality in multiplex networks

We next show that one should be careful when applying Bonacich centrality in multiplexed networks. In linear-quadratic utility function games played on simplex networks, utility and Bonacich centrality of agents are proportional (Ballester et al., 2006). In multiplexed networks, as we will show in this section, this is not always the case. We begin by providing some network definitions. We then show, by means of an example, that utility and Bonacich centrality do not necessarily go

hand in hand in multiplexed networks.

Network definitions. We define the *strength* s_i of node i as $s_i = \sum_{k=1}^K \sum_{j \neq i} w_{ij}^k$, which is a generalization of the notion of degree in weighted networks introduced by Barrat et al. (2004). We also define the row vector $\mathbf{S}$ in which entry i equals s_i . The Bonacich centrality of agent i in the multiplexed network is given by entry i of vector $\mathbf{BC}$,

$$\mathbf{BC} = (\mathbf{I}_N - b\mathbf{G}_M)^{-1}\mathbf{S}, \quad (12)$$

where $\mathbf{I}_N$ is the identity matrix of dimension N , and $0 < b < \frac{1}{\lambda_{\max}(\mathbf{G}_M)}$ is a parameter which is low enough so that Bonacich centrality takes a finite value. The definition in (12) should be interpreted as a simple and natural extension of Bonacich centrality to multiplexed networks, obtained by applying the standard measure to the aggregated adjacency matrix, and taking into account weights of links through vector $\mathbf{S}$. The Bonacich centrality measure in (12) is not the unique notion of centrality in multiplexed networks; rather, it serves as a convenient benchmark for illustrating how the relationship between centrality and utility can break down once interactions span multiple activities.

We next show by means of a simple example that utility and Bonacich centrality can both move in the same direction in multiplexed networks, or in opposite directions.

Example 5. Suppose that $c = 2$, $\alpha = 1$ and $b = 0.1$, and consider the preference parameters given in Table 4. Then,

$$\mathbf{T}_R = \mathbf{T}_S = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}, \mathbf{S} = \begin{bmatrix} 1.5 \\ 1.5 \\ 1.5 \\ 1.5 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{BC} = \begin{bmatrix} 3.75 \\ 3.75 \\ 3.75 \\ 3.75 \end{bmatrix}.$$

Suppose that $c = 2$, $\alpha = 1$, and $b = 0.1$, and consider the preference parameters given in Table 7. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.5 \\ 0.55 \\ 0.45 \\ 0.5 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.5 \\ 0.45 \\ 0.55 \\ 0.5 \end{bmatrix}, \mathbf{S} = \begin{bmatrix} 1.5 \\ 1.495 \\ 1.495 \\ 1.5 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1 \\ 1.01 \\ 1.01 \\ 1 \end{bmatrix} \text{ and } \mathbf{BC} = \begin{bmatrix} 3.746 \\ 3.742 \\ 3.742 \\ 3.746 \end{bmatrix}.$$

Suppose that $c = 2$, $\alpha = 1$, and $b = 0.1$, and consider the preference parameters given in Table 8. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.5 \\ 0.6 \\ 0.4 \\ 0.5 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.5 \\ 0.4 \\ 0.6 \\ 0.5 \end{bmatrix}, \mathbf{S} = \begin{bmatrix} 1.5 \\ 1.48 \\ 1.48 \\ 1.5 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1 \\ 1.04 \\ 1.04 \\ 1 \end{bmatrix} \text{ and } \mathbf{BC} = \begin{bmatrix} 3.73 \\ 3.72 \\ 3.72 \\ 3.73 \end{bmatrix}.$$

Suppose that $c = 2$, $\alpha = 1$, and $b = 0.1$, and consider the preference parameters given in Table 10. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.4 \\ 0.2 \\ 0.2 \\ 0.2 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.6 \\ 0.8 \\ 0.8 \\ 0.8 \end{bmatrix}, \mathbf{S} = \begin{bmatrix} 1.68 \\ 1.92 \\ 1.92 \\ 1.92 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1.04 \\ 1.36 \\ 1.36 \\ 1.36 \end{bmatrix} \text{ and } \mathbf{BC} = \begin{bmatrix} 4.5 \\ 4.7 \\ 4.7 \\ 4.7 \end{bmatrix}.$$

In the economies of tables 4, 7 and 8, which are all symmetric economies of heterogeneous preferences, the Bonacich centrality of an agent decreases whenever utility increases. In the economy of Table 10, which is an economy of heterogeneous preferences that is not symmetric, utility and Bonacich centrality of agents both increase. $\square$

The purpose of this example is not to suggest that Bonacich centrality, as defined in (12), is an appropriate measure of agents' performance or welfare in multiplexed networks. Rather, the example should be interpreted as a cautionary message. In simplex networks, Bonacich centrality is proportional to equilibrium utility, which often motivates its use as a proxy for agents' payoffs. In multiplexed networks, however, the different activities represent different types of interactions and cannot, in general, be treated as equivalent. As a result, even natural extensions of Bonacich centrality fail to preserve the monotonic relationship between centrality and utility. Treating multiplexed interactions as if they could be meaningfully summarized by a single centrality index can therefore be misleading.

4 Conclusion

Relationships where individuals interact in two or more activities have mainly been studied in fields other than Economics, and are called *multiplexed*. We study a game in which individuals allocate a limited amount of time between multiple activities, and form relationships in each of these activities. We provide a condition under which multiplexed networks arise, and show that networks that arise in equilibrium are not necessarily socially efficient. We then show that agents are better off when preferences, costs and strategic complementarity parameters are heterogeneous, than when they are homogeneous across activities. Intuitively, diversity allows for specialization: people spend most of their time on the activities they benefit the most from, leading to a higher overall welfare. We finally argue that one should be careful when applying Bonacich centrality in multiplexed networks, since the positive relationship between centrality and utility that holds in simplex networks does not necessarily carry over to multiplexed networks.

Appendix A Spreads that are not mean-preserving

When spreads are not mean-preserving, changes in utility are not only driven by a reallocation of preferences, but also by changes in the total time that agents allocate to activities. We provide a simple example that shows that utility can either increase or decrease, depending on whether the total time that agents allocate to activities increases or decreases. We consider two economies of heterogeneous preferences: one in which the utility of agents increases with respect to the economy of homogeneous preferences, in Table 11, and one in which the utility of agents decreases with respect to the economy of homogeneous preferences, in Table 12.

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 1$	$a_{2,R} = 1$	$a_{3,R} = 0.4$	$a_{4,R} = 0.4$
Activity S	$a_{1,S} = 0.4$	$a_{2,S} = 0.4$	$a_{3,S} = 1$	$a_{4,S} = 1$

Table 11: Heterogeneous preferences

Preference parameters				
	Agent 1	Agent 2	Agent 3	Agent 4
Activity R	$a_{1,R} = 0.6$	$a_{2,R} = 0.6$	$a_{3,R} = 0.1$	$a_{4,R} = 0.1$
Activity S	$a_{1,S} = 0.1$	$a_{2,S} = 0.1$	$a_{3,S} = 0.6$	$a_{4,S} = 0.6$

Table 12: Heterogeneous preferences

Example 6. Suppose that $c = 3$, $\alpha = 1$ and $b = 0.1$, and consider the preference parameters given in Table 4, which correspond to homogeneous preferences. Then,

$$\mathbf{T}_R = \mathbf{T}_S = \begin{bmatrix} 0.17 \\ 0.17 \\ 0.17 \\ 0.17 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 0.17 \\ 0.17 \\ 0.17 \\ 0.17 \end{bmatrix}$$

Suppose that $c = 3$, $\alpha = 1$ and $b = 0.1$, and consider the preference parameters given in Table 11. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.28 \\ 0.28 \\ 0.19 \\ 0.19 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.19 \\ 0.19 \\ 0.28 \\ 0.28 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 0.34 \\ 0.34 \\ 0.34 \\ 0.34 \end{bmatrix}$$

Suppose that $c = 3$, $\alpha = 1$ and $b = 0.1$, and consider the preference parameters given in Table 12. Then,

$$\mathbf{T}_R = \begin{bmatrix} 0.15 \\ 0.15 \\ 0.08 \\ 0.08 \end{bmatrix}, \mathbf{T}_S = \begin{bmatrix} 0.08 \\ 0.08 \\ 0.15 \\ 0.15 \end{bmatrix} \text{ and } \mathbf{U} = \begin{bmatrix} 0.09 \\ 0.09 \\ 0.09 \\ 0.09 \end{bmatrix}$$

When spreads in preferences are not mean-preserving, utility of agents can either increase or decrease. $\square$

Appendix B Proofs

Proof of Lemma 1. The first-order conditions of the utility function are:

$$\mathbf{A}_k^\top + \alpha \cdot \boldsymbol{\Psi} \cdot \mathbf{T}_k - 2c \cdot \mathbf{I}_N \cdot \mathbf{T}_k = \mathbf{0}.$$

Rewriting them yields:

$$\mathbf{A}_k^\top = 2c \cdot \mathbf{I}_N \cdot \mathbf{T}_k - \alpha \cdot \Psi \cdot \mathbf{T}_k,$$

$$\mathbf{T}_k = [2c \cdot \mathbf{I}_N - \alpha \cdot \Psi]^{-1} \mathbf{A}_k^\top.$$

Because $\sum_{k=1}^K \mathbf{T}_k \leq \mathbf{1}$, we have:

$$[2c \cdot \mathbf{I}_N - \alpha \cdot \Psi]^{-1} \sum_{k=1}^K \mathbf{A}_k^\top \leq \mathbf{1},$$

$$\sum_{k=1}^K \mathbf{A}_k^\top \leq [2c \cdot \mathbf{I}_N - \alpha \cdot \Psi] \mathbf{1}.$$

This concludes the proof of Lemma 1. $\square$

Proof of Lemma 2.

Prove part (i). Nash equilibrium values are given by $\mathbf{T}_k = [2c \cdot \mathbf{I}_N - \alpha \cdot \Psi]^{-1} \mathbf{A}_k^\top$, and all entries are of $[2c \cdot \mathbf{I}_N - \alpha \cdot \Psi]^{-1}$ are well-defined and strictly positive, given Assumption 1. It follows that entry i of $\mathbf{T}_k$ is strictly positive if and only if there is at least one agent $j \in \mathcal{N}$ such that $a_{j,k} > 0$. If $a_{i,k} = 0$, then $t_{i,k}^* > 0$ if and only if there is at least one agent $j \neq i$ such that $a_{j,k} > 0$.

Prove part (ii). We have shown in Step 1 that $t_{i,k}^* > 0$ if and only if there is at least one agent $j \in \mathcal{N}$ such that $a_{j,k} > 0$. It follows that there exist two activities k and l , with $k \neq l$, and two agents i and j , with $i \neq j$, such that $t_{i,k}^* > 0$, $t_{j,k}^* > 0$, $t_{i,l}^* > 0$ and $t_{j,l}^* > 0$ —i.e. the network is multiplexed—, if and only if $a_{i,k} > 0$ and $a_{j,l} > 0$ for any $i, j \in \mathcal{N}$. This concludes the proof of Lemma 2. $\square$

Proof of Proposition 1. Suppose that $\mathcal{E}$ is an economy of homogeneous preferences.

Prove part (i). Given Nash equilibrium values in (5), the total amount of time that each agent spends across activities, $\sum_{k=1}^K t_{i,k}^*$, is the same in $\mathcal{E}$ as in $\mathcal{E}'$. Notice that in $\mathcal{E}$, $t_{i,k}^* = \sum_{k=1}^K t_{i,k}^* / K$, from which it follows that $\sum_{k=1}^K t_{i,k}^{*2} = K \cdot (\sum_{k=1}^K t_{i,k}^* / K)^2 = K \cdot (\sum_{k=1}^K t_{i,k}^*)^2 / K^2 = (\sum_{k=1}^K t_{i,k}^*)^2 / K$. Therefore, inequal-

ity $(\sum_{k=1}^K t_{i,k}^*)^2/K \leq \sum_{k=1}^K t_{i,k}^{*2}$ for any $\sum_{k=1}^K t_{i,k}^*$ implies that $\sum_{k=1}^K t_{i,k}^{*2}$ is weakly larger for all $i \in \mathcal{N}$ in $\mathcal{E}'$ than in $\mathcal{E}$. The previous inequality is precisely Jensen's inequality (Jensen, 1906), and thus, holds. Since $U_i = c \sum_{k=1}^K t_{i,k}^{*2}$ at equilibrium, we conclude $U_i^{*\mathcal{E}'} \geq U_i^{*\mathcal{E}}$ for all $i \in \mathcal{N}$.

Prove part (ii). From the Nash equilibrium values given in (5), it follows immediately that (i) in the economy of homogeneous preferences $\mathcal{E}$, $t_{i,k}^* = t_{j,l}^* = t$ for all $i, j \in \mathcal{N}$ and all $k, l \in \mathcal{K}$, and that (ii) in a symmetric economy of heterogeneous preferences, $\mathcal{N}$ can be partitioned into three disjoint sets $\mathcal{N}_1$, $\mathcal{N}_2$ and $\mathcal{N}_3$, where $|\mathcal{N}_1| = |\mathcal{N}_2|$, such that there is exactly one agent $j \in \mathcal{N}_2$ for each agent $i \in \mathcal{N}_1$, such that for this pair of agents $i, j \in \mathcal{N}$, it holds that $t_{i,k}^* = t + x_{i,k}$ for $k \in \{1, \dots, f\}$, $t_{i,k}^* = t - x_{i,k}$ for $k \in \{f+1, \dots, g\}$ and $t_{i,k}^* = t$ for $k \in \{g+1, \dots, K\}$, and $t_{j,k}^* = t - x_{i,k}$ for $k \in \{1, \dots, f\}$, $t_{j,k}^* = t + x_{i,k}$ for $k \in \{f+1, \dots, g\}$ and $t_{j,k}^* = t$ for $k \in \{g+1, \dots, K\}$ for some f and g , and such that for all $i \in \mathcal{N}_3$, $t_{i,k}^* = t$ for all $k \in \{1, \dots, K\}$, where $\sum_{k=1}^K t_{i,k}^* = Kt$ for all $i \in \mathcal{N}$ and $x_{i,k} > 0$ for all $i \in \mathcal{N}$ and $k \in \mathcal{K}$, and that (iii) every $x_{i,k}$ is weakly larger in $\mathcal{E}''$ than in $\mathcal{E}'$. Given that the utility of each agent i is $U_i = c \sum_{k=1}^K t_{i,k}^{*2}$ at equilibrium, and thus that $\frac{\partial U_i^*}{\partial t_{i,k}^*} > 0$ and $\frac{\partial^2 U_i^*}{\partial t_{i,k}^{*2}} > 0$ for any $k \in \mathcal{K}$, U_i^* is weakly larger in $\mathcal{E}''$ than in $\mathcal{E}'$ for all $i \in \mathcal{N}$. This concludes the proof of Proposition 1. $\square$

Proof of Proposition 2. Suppose that $\mathcal{E}$ is an economy of homogeneous parameters and that $\mathcal{E}''$ and $\mathcal{E}'$ are two economies of heterogeneous parameters (α_k or c_k) such that $\mathcal{E}''$ has a larger spread than $\mathcal{E}'$ with respect to $\mathcal{E}$. The Nash equilibrium value of agent i in activity k , provided that the utility function is the one given in (6), is given by entry i of $\mathbf{T}_k = [2c_k \cdot \mathbf{I}_N - \alpha_k \cdot \Psi]^{-1} \mathbf{A}_k^\top$. By expansion of the Neumann series, $[2c_k \cdot \mathbf{I}_N - \alpha_k \cdot \Psi]^{-1} = \frac{1}{2c_k} \sum_{n=0}^{\infty} (\frac{\alpha_k \Psi}{2c_k})^n$. The utility of agent i at equilibrium is therefore given by row i of $\mathbf{U} = \sum_{k=1}^K c_k [\frac{1}{2c_k} \sum_{n=0}^{\infty} (\frac{\alpha_k \Psi}{2c_k})^n] \mathbf{A}_k^\top \circ [\frac{1}{2c_k} \sum_{n=0}^{\infty} (\frac{\alpha_k \Psi}{2c_k})^n] \mathbf{A}_k^\top = \sum_{k=1}^K \frac{1}{4c_k} [\sum_{n=0}^{\infty} (\frac{\alpha_k \Psi}{2c_k})^n] \mathbf{A}_k^\top \circ [\sum_{n=0}^{\infty} (\frac{\alpha_k \Psi}{2c_k})^n] \mathbf{A}_k^\top$, where $\circ$ is the Hadamard product. Since $\frac{\partial \mathbf{U}}{\partial \alpha_k} > \mathbf{0}$, $\frac{\partial^2 \mathbf{U}}{\partial \alpha_k^2} > \mathbf{0}$, $\frac{\partial \mathbf{U}}{\partial c_k} < \mathbf{0}$ and $\frac{\partial^2 \mathbf{U}}{\partial c_k^2} > \mathbf{0}$ for any $k \in \mathcal{K}$, we conclude that $U_i^{*\mathcal{E}'} > U_i^{*\mathcal{E}}$ and $U_i^{*\mathcal{E}''} > U_i^{*\mathcal{E}'}$ for all $i \in \mathcal{N}$. This concludes the proof of Proposition 2. $\square$

Proof of Proposition 3. Consider an economy $\mathcal{A}$, in which every agent i spends

a strictly positive amount of time on only one activity each, and a total amount of time given by row i of $\bar{\mathbf{T}}$. Consider as well an economy $\mathcal{B}$, in which every agent i spends a total amount of time given by row i of $\bar{\mathbf{T}}$, and in which every agent spends a strictly positive amount of time on each activity. We denote by $\sum_{i=1}^N U_i^A$ the sum of utilities in economy $\mathcal{A}$, and by $\sum_{i=1}^N U_i^B$ the sum of utilities in economy $\mathcal{B}$. Consider any set of parameters. Increase c by some amount $\varepsilon > 0$. Partial derivative $\frac{\partial c \sum_{i=1}^N \sum_{k=1}^K t_{i,k}^2}{\partial c}$ is larger in economy $\mathcal{A}$ than in economy $\mathcal{B}$. Therefore, difference $\sum_{i=1}^N U_i^A - \sum_{i=1}^N U_i^B$ decreases, and either stays positive, becomes zero, becomes negative, or stays negative. Suppose first that it either becomes or stays negative. Then, there exists some $\underline{c}$ such that, if $c > \underline{c}$, then $\sum_{i=1}^N U_i$ is strictly larger if every agent spends a strictly positive amount of time on each activity, than if every agent spends a strictly positive amount of time on only one activity each—i.e., $\sum_{i=1}^N U_i$ is strictly larger if the network is multiplexed. Since parameters are chosen without loss of generality, $\underline{c}$ exists given any set of parameters. Suppose next that $\sum_{i=1}^N U_i^A - \sum_{i=1}^N U_i^B$ either stays positive or becomes zero. Note that, if c increases by ε at multiple iterations, then the difference $\sum_{i=1}^N U_i^A - \sum_{i=1}^N U_i^B$ decreases by the same amount at each iteration. By induction of the argument, increasing c by amount $\varepsilon > 0$ multiple times leads $\sum_{i=1}^N U_i^A - \sum_{i=1}^N U_i^B$ to become negative at some iteration. Therefore, there exists some $\underline{c}$ such that, if $c > \underline{c}$, then $\sum_{i=1}^N U_i$ is strictly larger if every agent spends a strictly positive amount of time on each activity, than if every agent spends a strictly positive amount of time on only one activity each, i.e. $\sum_{i=1}^N U_i$ is strictly larger if the network is multiplexed. Since parameters are chosen without loss of generality, $\underline{c}$ exists given any set of parameters. This concludes the proof of Proposition 3. $\square$

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